MGM Cancer Institute

The Department of Paediatric Haematology, Oncology, Blood and Marrow Transplantation is a speciality clinic at MGM Cancer Institute – Chennai. It is spearheaded by Dr M. Deenadayalan who, along with his team, treats a variety of children's illnesses ranging from benign and malignant haematology, solid tumours, as well as performing Blood & Marrow Transplantation (BMT).

According to WHO data nearly four lakh children are diagnosed with cancer every year throughout the world, with 50,000 of these cases from India alone. Over the last two to three decades, the incidence of cancer detection has gradually gone up, but this is likely due to increased awareness and early diagnosis. Hearteningly, the cure rate has improved due to advances in chemotherapy, better supportive care, surgical techniques, radiotherapy and newer forms of treatments like stem cell transplants (also called Blood and Marrow transplants), immunotherapy and cellular therapy.

The Department of Paediatric Haematology, Oncology, Blood and Marrow Transplantation at MGM Cancer Institute – Chennai is well equipped to offer these cutting-edge therapies and handle the unique challenges of treating children. The Department is managed by a team of three doctors—Dr M. Deenadayalan, HOD and Clinical Lead; Dr Vimal Kumar, Senior Consultant; and Dr Rishab Bharadwaj, Consultant. Dr M. Deenadayalan is an esteemed Paediatric Haematologist and Oncologist known for his extensive experience in treating a range of paediatric conditions. With over 18 years in the field, he specialises in benign blood disorders, malignant haematology, immunodeficiencies and solid tumours such as neuroblastoma, germ cell tumours, sarcoma, hepatoblastoma, and brain tumours.



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AN ADVANCED CENTRE FOR TREATING CHILDREN

The Department is equipped to handle three verticals: paediatric haematology, oncology and BMT. They cover a wide spectrum of ailments, from benign blood disorders like nutritional anaemia, thalassemia, sickle-cell anaemia to blood cancers like leukaemias and lymphomas, as well as solid tumours. A number of these conditions may require Blood & Marrow Transplantation (BMT), otherwise known as Haematopoietic Stem Cell Transplantation (HSCT), where the patient's diseased bone marrow is replaced with healthy stem cells.

As a dedicated cancer treatment facility, MGM Cancer Institute (a unit of MGM Healthcare), is fully equipped to address paediatric concerns, ably supported by the other specialities in the hospital. For instance, for paediatric oncology treatment, the hospital has trained paediatric trained staff to administer chemotherapy, a dedicated chemotherapy daycare centre, 24x7 Paediatric Intensive Care (PICU) support, and a highly advanced laboratory for all the unique and diverse investigations needed. They are also supported by a well-equipped blood bank, as these patients require a lot of specialised blood products. The blood bank also boasts of the latest technology with apheresis machines which aid in collecting peripheral blood stem cells from donors. As a quaternary health care centre, it is also one of the few hospitals with all departments under one roof such as Radiotherapy, Nuclear Medicine (PET-CT) and various Surgical specialities.

"A bone marrow transplant is one of the most specialised and highly skilled procedures in Medicine. MGM Cancer Institute has a state-of-the-art, six-bedded Blood & Marrow Transplantation (BMT) unit with HEPA-filtered and positive-pressure rooms. It is an excellent setup to ensure complete sterility," says Dr M. Deenadayalan.

ADVANCES IN BLOOD & MARROW TRANSPLANTATION (BMT)

At MGM Cancer Institute, the Department performs two types of Blood & Marrow Transplantation (BMT): autologous, where the patient's own stem cells are collected prior to chemotherapy and infused later; as well as the more complex allogeneic stem cell transplants, where the abnormal stem cells of the patient are replaced with those from a healthy donor. "BMT is the only permanently curative option for conditions like thalassemia, sickle-cell anaemia, relapsed leukaemias, immunodeficiencies, bone marrow failure syndromes as well as some inherited metabolic disorders", informs Dr. Deenadayalan.

For allogeneic transplants, donors are identified using high-resolution HLA (human leukocyte antigen) typing. Finding a full match can be challenging, which is where half-matched or haploidentical transplant comes to the rescue. "Using the advanced alpha/beta depletion technique, we are able to perform highly complex half-matched transplants. MGM Cancer Institute is one of the few centres offering this sophisticated procedure", says Dr M. Deenadayalan.

The Department's most recent transplant was for a genetic condition called Bloom Syndrome, which was complicated by the presence of Myelodysplastic Syndrome (MDS), as both lie at the opposite ends of the spectrum. The donor was the younger sibling and only a half-matched donor, so special care had to be taken to harvest the stem cells. Using all their experience, the procedure was pulled off successfully and the patient is recovering well.

MGM CANCER INSTITUTE
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